

Exercise 1: Configuring a Basic Spring Application

MainApp.java

```
package com.library

import org.springframework.context.ApplicationContext;
import
org.springframework.context.support.ClassPathXmlApplicationConte
xt;

import com.library.repository.BookRepository;
import com.library.service.BookService;

public class MainApp {

    public static void main(String[] args) {

        ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =

            context.getBean("bookService", BookService.class)

        BookRepository repository =

            context.getBean("bookRepository", BookRepository.class);

        service.displayService();

        repository.displayRepository();

    }

}
```

```
LibraryManagement > src > main > java > com > library > J MainApp.java > MainApp
1 package com.library;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 import com.library.repository.BookRepository;
7 import com.library.service.BookService;
8
9 public class MainApp {
10
11     public static void main(String[] args) {
12
13         ApplicationContext context =
14             new ClassPathXmlApplicationContext("applicationContext.xml");
15
16         BookService service =
17             context.getBean("bookService", BookService.class);
18
19         BookRepository repository =
20             context.getBean("bookRepository", BookRepository.class);
21
22         service.displayService();
23     }
24 }
```

PS C:\Users\karun\OneDrive\Desktop\cognizant programs> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.5.11-hotspot\bin\java.exe' '@C:\Users\karun\AppData\Local\Temp\cp_c23kc71rdliwpyihpr906' jls.argsfile' com.library.MainApp
Book Service is working.
Book Repository is working.
PS C:\Users\karun\OneDrive\Desktop\cognizant programs>

Exercise 2: Implementing Dependency Injection

MainApp.java

```
package com.library;
```

```
import org.springframework.context.ApplicationContext;
```

```
import
```

```
org.springframework.context.support.ClassPathXmlApplicationConte  
xt;
```

```
import com.library.service.BookService;
```

```
public class MainApp {
```

```
    public static void main(String[] args) {
```

```
        ApplicationContext context =
```

```
            new
```

```
ClassPathXmlApplicationContext("applicationContext.xml");
```

```

        BookService service =context.getBean("bookService",
BookService.class);

        service.displayService();

    }

}

```

```

J MainApp.java 1 X
LibraryManagement > src > main > java > com > library > J MainApp.java > ...
3  import org.springframework.context.ApplicationContext;
4  import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6  import com.library.service.BookService;
7
8  public class MainApp {
9
10     Run | Debug
11     public static void main(String[] args) {
12
13         ApplicationContext context =
14             new ClassPathXmlApplicationContext("applicationContext.xml");
15
16         BookService service =
17             context.getBean("bookService", BookService.class);
18
19         service.displayService();
20     }
21 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR

```

PS C:\Users\karun\OneDrive\Desktop\cognizant programs> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.5.11-hotspot\bin\java.exe' '@C:\Users\karun\AppData\Local\Temp\cp_cz3kc7lrdiiwpyihpr906
jla.argsfile' 'com.library.MainApp'
Book Service is working.
Book Repository is working.
PS C:\Users\karun\OneDrive\Desktop\cognizant programs>

```

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Exercise 4: Creating and Configuring a Maven Project

pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
```

```
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
```

```
    http://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.library</groupId>
```

```
<artifactId>LibraryManagement</artifactId>
<version>1.0-SNAPSHOT</version>
<properties>
  <maven.compiler.source>1.8</maven.compiler.source>
  <maven.compiler.target>1.8</maven.compiler.target>
  <project.build.sourceEncoding>UTF-
8</project.build.sourceEncoding>
</properties>
<dependencies>
  <!-- Spring Context -->
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.3.30</version>
  </dependency>
  <!-- Spring AOP -->
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-aop</artifactId>
    <version>5.3.30</version>
  </dependency>
  <!-- Spring Web MVC -->
  <dependency>
    <groupId>org.springframework</groupId>
```

```

<artifactId>spring-webmvc</artifactId>

<version>5.3.30</version>

</dependency>

</dependencies>

<build>

  <plugins>

    <plugin>

      <groupId>org.apache.maven.plugins</groupId>

      <artifactId>maven-compiler-plugin</artifactId>

      <version>3.13.0</version>

      <configuration>

        <source>1.8</source>

        <target>1.8</target>

      </configuration>

    </plugin>

  </plugins>

</build>

</project>

```

